

6 – EXERCISES ON MAIN AND GLOBAL VS LOCAL VARIABLES

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Quick reference

Local vs. global variables

```
int global_volume = 80; // global: exists for the entire program,
                        // visible to every function
void print_volume() {
    int local_boost = 10; // local: exists only inside this function
    std::cout << global_volume + local_boost << std::endl;
} // local_boost is destroyed here; global_volume lives on
```

Key rules:

- A local variable **shadows** a global with the same name inside its scope.
- Modifying a global inside a function affects all subsequent callers.
- Prefer local variables; globals make programs harder to reason about.

Passing arguments to main

Remember a program can be executed from a terminal / command-line. To do that, we write:

`./name_of_the_program.`

In Compiler Explorer / Godbolt, the program is always named `./output.s.`

We can pass arguments to the program which will be passed to main.

```
int main(int argc, char* argv[]) {
    // argc : number of arguments (including the program name)
    // argv : sequence of strings; argv[0] is the program name,
    //       argv[1] is the first user-supplied argument, etc.
}
```

For example, on the command line we could write:

`./my_program hello 42`

And in that case:

```
// argc == 3
// argv[0] contains "./my_program"
// argv[1] contains "hello"
// argv[2] contains "42"
```

Reading string arguments

`argv` contains variables of type `char *`, which we don't know how to handle yet.

To manipulate string arguments contained in `argv`, you must first store it in a `std::string` like so:

```
1 std::string str_arg = argv[1];
```

You can also convert it if you want to do comparison on arguments without having to store them:

```
1 if(std::string(argv[1]) == "a_string"){
2     // do something...
3 }
```

`atoi` — converting a string argument to an integer

```
#include <cstdlib> // required for atoi
```

```
int n = atoi(argv[1]); // converts the string "42" to the int 42
```

`atoi` returns 0 if the string cannot be converted (e.g. `atoi("hello")== 0`). It does *not* signal an error — always make sure `argc` is large enough before reading `argv[i]`.

Part A — Local vs. global variables

Problem 1 — Trace global state

What does this program print?

Global volume

```
1  #include <iostream>
2
3  int volume = 50;
4
5  void increase() {
6      volume = volume + 10;
7  }
8
9  void reset() {
10     volume = 0;
11 }
12
13 int main() {
14     std::cout << volume << std::endl; // line A
15     increase();
16     std::cout << volume << std::endl; // line B
17     increase();
18     increase();
19     std::cout << volume << std::endl; // line C
20     reset();
21     std::cout << volume << std::endl; // line D
22     return 0;
23 }
```

Line A:

Line B:

Line C:

Line D:

Problem 2 — Shadowing

What does this program print? Pay attention to which `colour` is used at each point.

Shadowing

```
1  #include <iostream>
2
3  int colour = 100;
4  void paint() {
```

```

5     int colour = 200;
6     std::cout << colour << std::endl; // line A
7 }
8 void tint() {
9     colour = colour + 50;
10    std::cout << colour << std::endl; // line B
11 }
12 int main() {
13     std::cout << colour << std::endl; // line C
14     paint();
15     std::cout << colour << std::endl; // line D
16     tint();
17     std::cout << colour << std::endl; // line E
18     return 0;
19 }

```

Line A:

Line B:

Line C:

Line D:

Line E:

Problem 3 — Scope lifetime

Does this program compile? If not, identify the line and explain why.

Scope lifetime

```

1  #include <iostream>
2
3  void set_hue(int h) {
4      int hue = h * 2;
5  }
6
7  int main() {
8      set_hue(45);
9      std::cout << hue << std::endl; // line X
10     return 0;
11 }

```

Does it compile?

Why:

Problem 4 — Write the code

Complete this exercise here: <https://godbolt.org/z/K99f4YEb4>.

You have to:

- Complete `void print_hue()` that prints the current value of `hue`.
- Complete the function `void shift_hue(int amount)`: it adds `amount` to the global `hue`. Then, if `hue` is outside the range `[0, 360]`, it should set it back to 0 if it inferior to zero, or 360 if it was superior (strictly) to 360.
- In `main`, write the following calls:

```
- shift_hue(200)
- print_hue()
- shift_hue(250)
- print_hue()
- shift_hue(-100)
- print_hue()
```

Expected output:

```
200
360
260
```

Part B — Arguments to `main`

Problem 1 — Reading `argc` and `argv`

The program below is invoked as:

```
./player Beethoven 120
```

What does it print?

Argument reader

```
1  #include <iostream>
2  #include <cstdlib>
3
4  int main(int argc, char* argv[]) {
5      std::cout << argc << std::endl; // line A
6      std::cout << argv[0] << std::endl; // line B
7      std::cout << argv[1] << std::endl; // line C
8      std::cout << argv[2] << std::endl; // line D
9      std::cout << atoi(argv[2]) + 10 << std::endl; // line E
10     return 0;
11 }
```

Line A:

Line B:

Line C:

Line D:

Line E:

Problem 2 — Argument count guard

What does this program print for each invocation below?

Argument guard

```
1  #include <iostream>
2  #include <cstdlib>
3
4  int main(int argc, char* argv[]) {
5      if (argc < 2) {
6          std::cout << "usage: ./filter <value>" << std::endl;
7          return 1;
8      }
9
10     int value = atoi(argv[1]);
```

```

11
12     if (value < 0 || value > 255) {
13         std::cout << "invalid" << std::endl;
14         return 1;
15     }
16
17     std::cout << "brightness: " << value << std::endl;
18     return 0;
19 }

```

`./filter` $\Rightarrow$

`./filter 200` $\Rightarrow$

`./filter 300` $\Rightarrow$

`./filter hello` $\Rightarrow$

What does `atoi("hello")` return, and what does the program therefore print for that call? .

Problem 3 — Spot the bug

This program should print the sum of all integer arguments passed to it. For example, `./sum 10 20 30` should print 60. It contains bugs.

Buggy argument sum

```

1  #include <iostream>
2  #include <cstdlib>
3  int main(int argc, char* argv[]) {
4      int total = 0;
5      for (int i = 0; i <= argc; i = i + 1) {
6          total = total + atoi(argv[i]);
7      }
8      std::cout << total << std::endl;
9      return 0;
10 }

```

Bug:

Fix:

Hint: what is `argv[0]`, and what does `atoi` return for it? What happens when `i == argc`?

Problem 4 — Write the code

This exercise should be completed here: <https://godbolt.org/z/1b47b45qh>. The program currently checks the amount of argument given to it. If it is not at least 3, it prints a message to

help the user understand how the program should be used (remember, the name of the program is already an argument, so `argc` is always at least 1).

Complete the program so that it prints `"<artist> performing at <bpm> bpm"`.

Then, change the program so that it now takes a 3rd user-provided argument: a tempo label. Your change should make it so that the program outputs the usage message if `argc` is not 4. And you should store the 3rd argument in a variable of type `std::string` then print it on a new line.

Expected output for `./output.s 90 Miles medium`:

```
Miles performing at 90 bpm
medium
```

Correction link: <https://godbolt.org/z/P1KGcKdfb>

Part C — Mixing everything

Problem 1 — Command-line image filter

This exercise has to be completed here: <https://godbolt.org/z/fnr83nb4q>. The goal of this exercise is to write a program that takes a command (`print`, `boost`, `reduce`) to control a variable containing a volume value. Complete the exercise by doing the following:

1. Complete the function `print`: it should be the value of the global variable `volume`.
2. Complete the function `restrain_value`: it returns a value that is either equal to `value` if it is in the range `[min, max]`, or equal to `min` if it is under, or equal to `max` otherwise.
3. Complete the function `multiply_gain`: it returns the result of the multiplication `gain × boost`, restrained in the range `[0, 100]`. *Hint: use the function `restrain_value` to restrain the result of the multiplication.*
4. Complete the function `subtract_from_gain`: it returns the result of the subtraction `gain – value`, restrained in the range `[0, 100]`. *Hint: use the function `restrain_value` to restrain the result of the multiplication.*
5. Complete the function `main`:
 - (a) Complete the `if-else if-else` checking for the value of `command`. The accepted values are: `"print"`, `"boost"`, `"reduce"`.
 - (b) For the case `"print"`, call the function `print_gain`
 - (c) For the case `"boost"`, call the function `multiply_gain` by passing it the variables `volume` and `value` and store the result in `volume`. Then, call `print_gain`
 - (d) For the case `"reduce"`, call the function `subtract_from_gain` by passing it the variables `volume` and `value` and store the result in `volume`. Then, call `print_gain`

Correction link here: <https://godbolt.org/z/3zMrhs6f5>